Unit 3 · Day 1

Atom Discovery: Democritus to Dalton

Name: _____ Period: _____ Date: _____

1. Cold Open (before the sim)

Mr. D tears a piece of foil in half, again and again. Circle your prediction, then give a reason.

I think a piece of matter can be cut in half forever / until I reach a piece that can't be cut

Because _____

_____

Gold prediction (Part 1 of the sim): number of halvings from 1 cm to one atom = _____ Actual: _____

2. Sim Data (fill in as you go)

Part 2, Make Water. Record three different trials.

Trial	H in (g)	O in (g)	Water (g)	Left over (what, how much)	Mass in = mass out?	O used ÷ H used
1						
2						
3						

Part 3, Two Compounds. Same hydrogen, different compounds.

Compound	H used (g)	O used (g)	O per 1 g of H (g)
A (water)			
B (hydrogen peroxide)			

O per 1 g H in B ÷ O per 1 g H in A = _____ Why a whole number? _____

3. Takeaways

Two thinkers. Fill the table.

	Big idea	What backed it up?
Democritus (≈400 BCE)		
Dalton (1803)		

Dalton's five postulates. Use the word bank. *Word bank: atoms · created · identical · rearranged · whole-number · subdivided*

1. All matter is composed of extremely small particles called _____.
2. Atoms cannot be _____, _____, or destroyed.
3. Atoms of a given element are _____ in size, mass and other properties. Atoms of different elements differ.
4. Atoms of different elements combine in simple _____ ratios to form compounds.
5. In chemical reactions, atoms are combined, separated, or _____.

Three laws. Name each one and give your example from the sim.

Idea	Law (name it)	My evidence from the sim
Total mass before = total mass after		
One compound always has the same mass recipe		
Two compounds of the same elements: recipes differ by simple whole-number ratios		

★ CHALLENGE TAKEAWAYS (if you got there)

Dalton assumed water = HO. If that is true, one O atom is _____ times as heavy as one H atom. If water is H_2O, it is _____ times.

Nitrogen oxides, O per 14 g N: _____ g, _____ g, _____ g. Simple ratio: ___ : ___ : ___ Law: _____

Beyond Dalton. Something a solid ball can't explain: _____

_____

4. Exit Ticket

One thing Dalton's model explains, and one thing it can't. Use the sentence: "Dalton's model explains ____ because ____, but it can't explain ____."

_____

_____

Rubric (10 points)	Points	What earns it
Cold open and gold prediction	2	Prediction circled with a reason; gold numbers filled in
Sim data tables	2	Trials and compound table filled with units
Two thinkers and five postulates	3	Table complete; postulates correct
Three laws and evidence	2	Each law named; each example comes from the sim
Exit ticket	1	Uses the sentence frame; one real explanation and one real limit
Stamp screen shown to Mr. D	—	Required to turn in the sheet. Challenge is extra, not required.